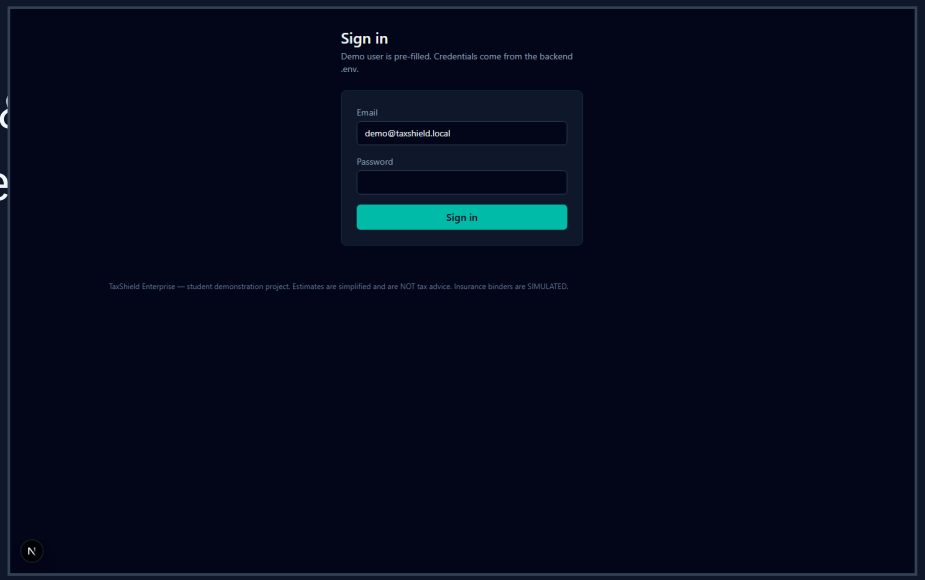
TaxShield Enterprise

Turn operational telemetry into a defensible R
— with every number traceable to an encrypte
ledger.



Team Qubrix · INTELLICON '26 Buildathon · student demonstration project

Not tax advice · insurance simulated · risk model trained on synthetic data

The problem

- **Mid-market digital firms leave R&D credits unclaimed.** Section 41 credits reward technological experimentation — but the evidence lives in Git history and cloud bills, not in tax workflows.
- **The evidence exists — in the wrong systems.** Commit messages, GPU training runs, SageMaker experiments are exactly what the IRS four-part test asks for, and nobody converts them into tax-ready documentation.
- **Nexus compounds the risk.** Post-2018 economic-nexus rules ($500k CA, $100k WA…) caught many multi-state SaaS firms unregistered.
- Gate 1 sized the market at $26.39B TAM (secondary sources, directional). A 50-person firm typically forgoes $50k–$250k per year.

Sources: IRS Form 6765 instructions and §41 Audit Technique Guide; per-state revenue departments (URLs in repo).

Who has it

Target customer: US software firms, 10–250 employees, $2M–$50M ARR, multi-state, on AWS + GitHub + Stripe, without a full-time credit specialist. Buyer: CFO/VP Finance.

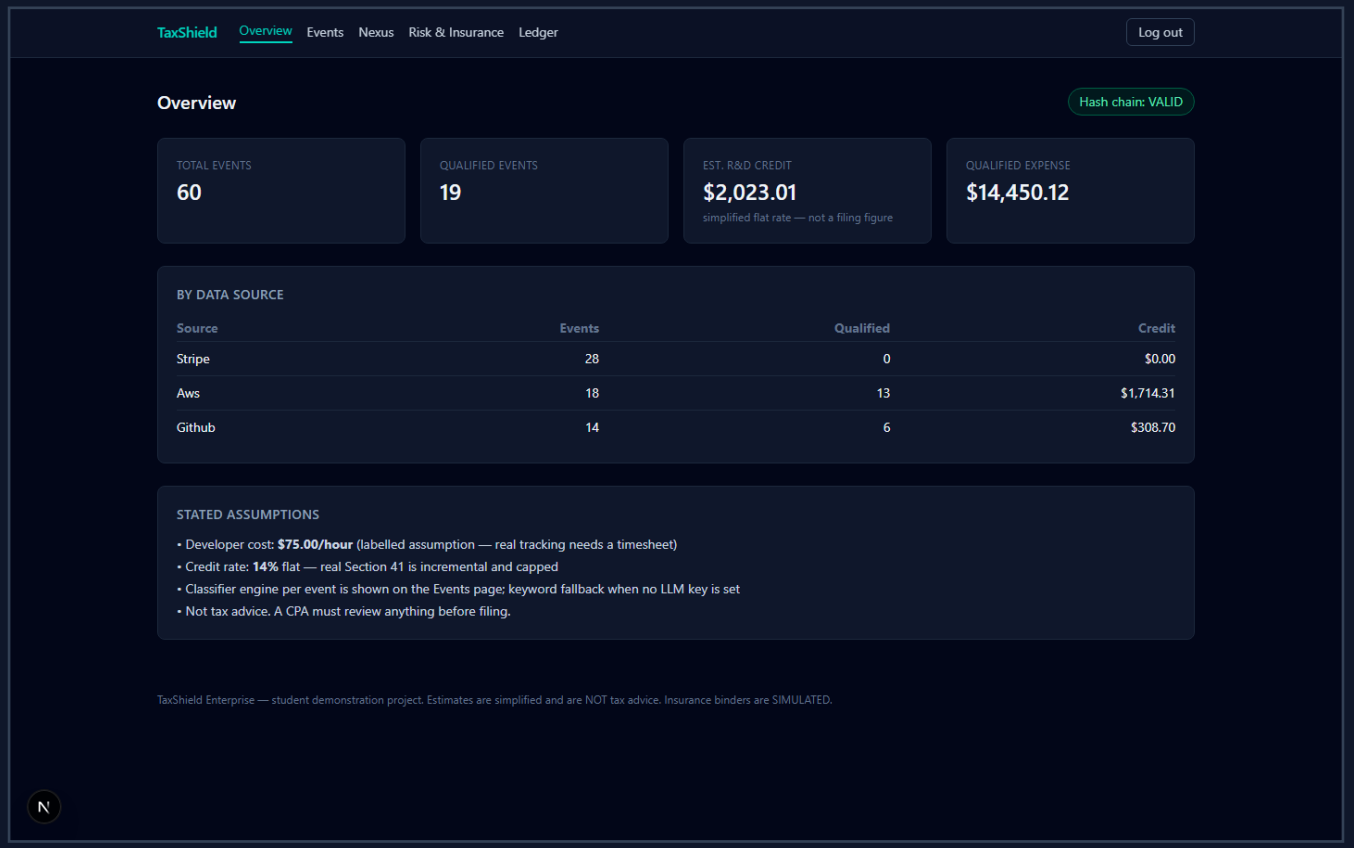
- Typical profile: 50 engineers, ~$12M ARR, 8 states of sales tax exposure, $2–4M/year of qualifying wages never mapped to the four-part test.
- Existing tools miss it: CPAs see spreadsheets, not telemetry; Avalara covers nexus, not credits; DIY software fills forms without evidence.

Honest disclosure: the 5+ customer interviews promised in Gate 1 have not been held yet. Interview script is ready; findings will revise this slide. We report the gap rather than invent quotes.

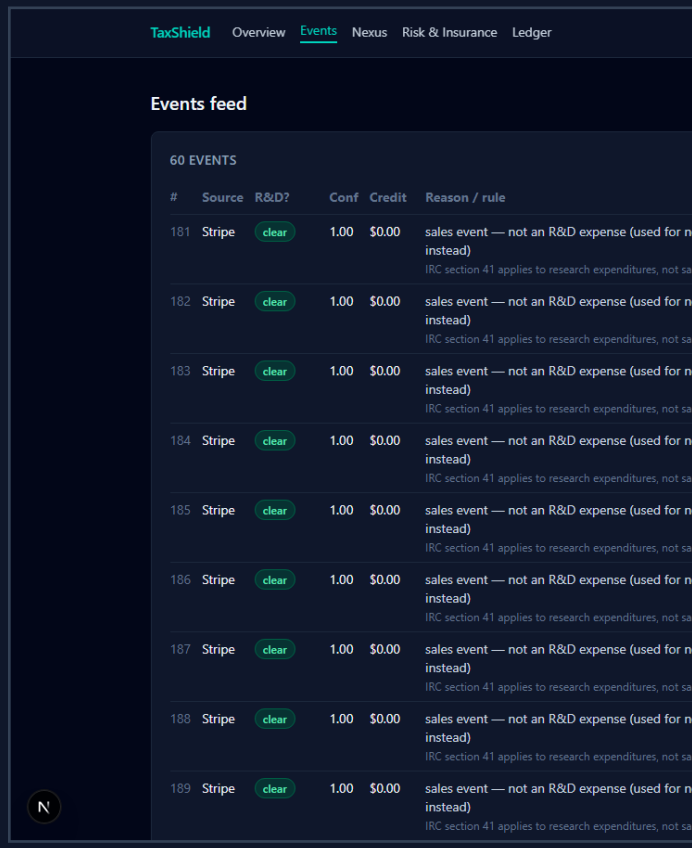
Solution

- **Ingest** GitHub commits, Stripe test-mode payments, AWS billing CSVs (or the built-in simulator for offline demos).
- **Classify** every event against official IRS §41 text: retrieval over 241 chunks from the Form 6765 instructions + Audit Technique Guide, LLM verdict in strict JSON, explainable keyword fallback. Engine named on every result.
- **Encrypt & chain:** AES-256-GCM per record, SHA-256 hash chain — tamper with one row and verification names the exact record.
- **Nexus watch** across 10 states with thresholds sourced from official revenue sites; unverified rows are flagged, not faked.
- **Risk score + simulated binder** with transparent formulas, labelled everywhere they appear.

Live demo — the dashboard

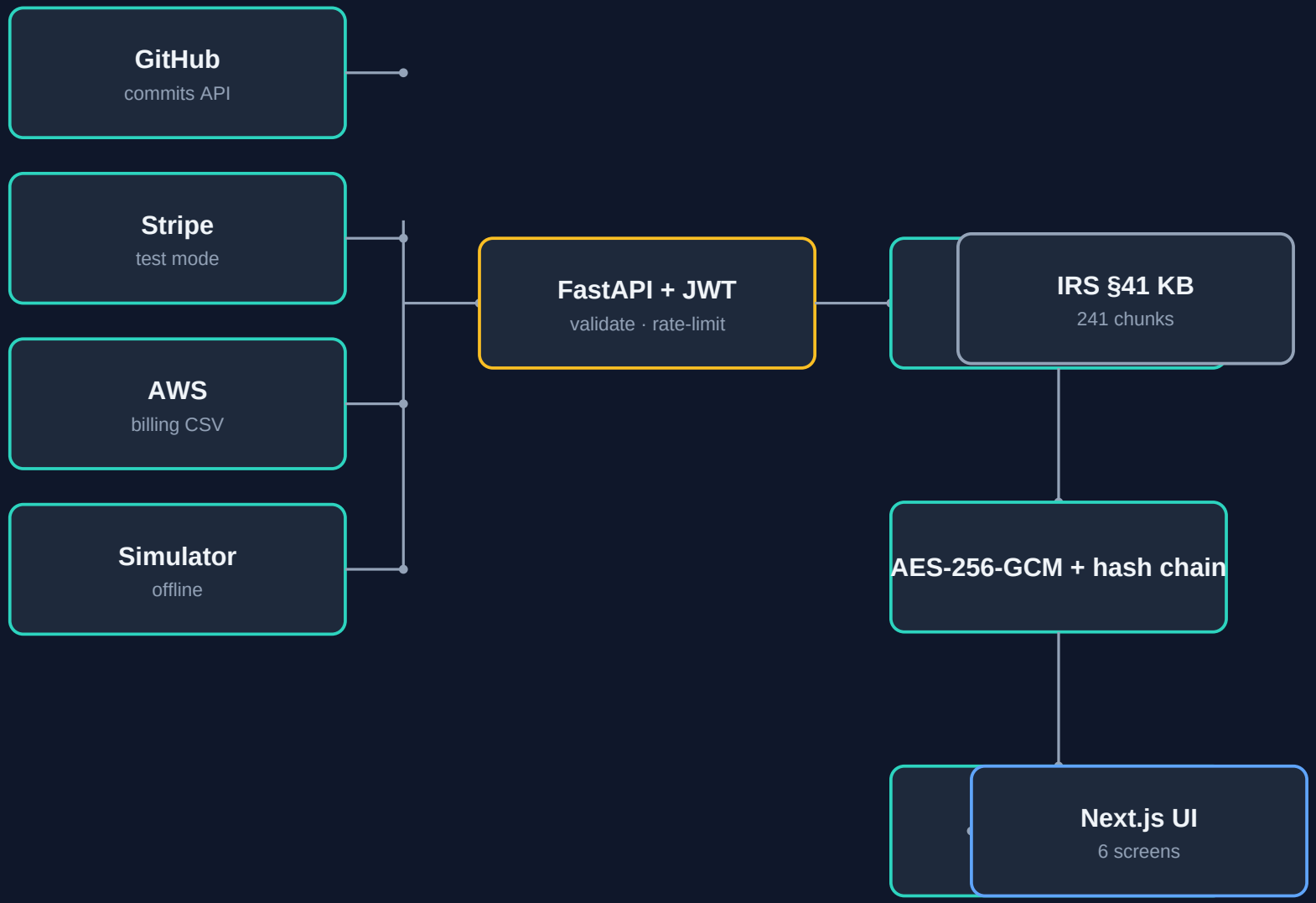


Overview: live totals + chain badge



Events: R&D verdict, confidence, IRS rule citation, er

How it works


GitHub
commits API
Stripe
test mode
AWS
billing CSV
Simulator
offline
FastAPI + JWT
validate · rate-limit
IRS §41 KB
241 chunks
AES-256-GCM + hash chain
Next.js UI
6 screens

AI approach & measured accuracy

Design: retrieve the 3–4 most relevant official IRS rule chunks (TF-IDF, offline, free), ask the LLM for one strict JSON verdict (qualified, confidence, reason, rule_cited, expense), parse safely, fall back to explainable keyword rules on ANY failure. Results cached — repeat demos never re-hit an API.

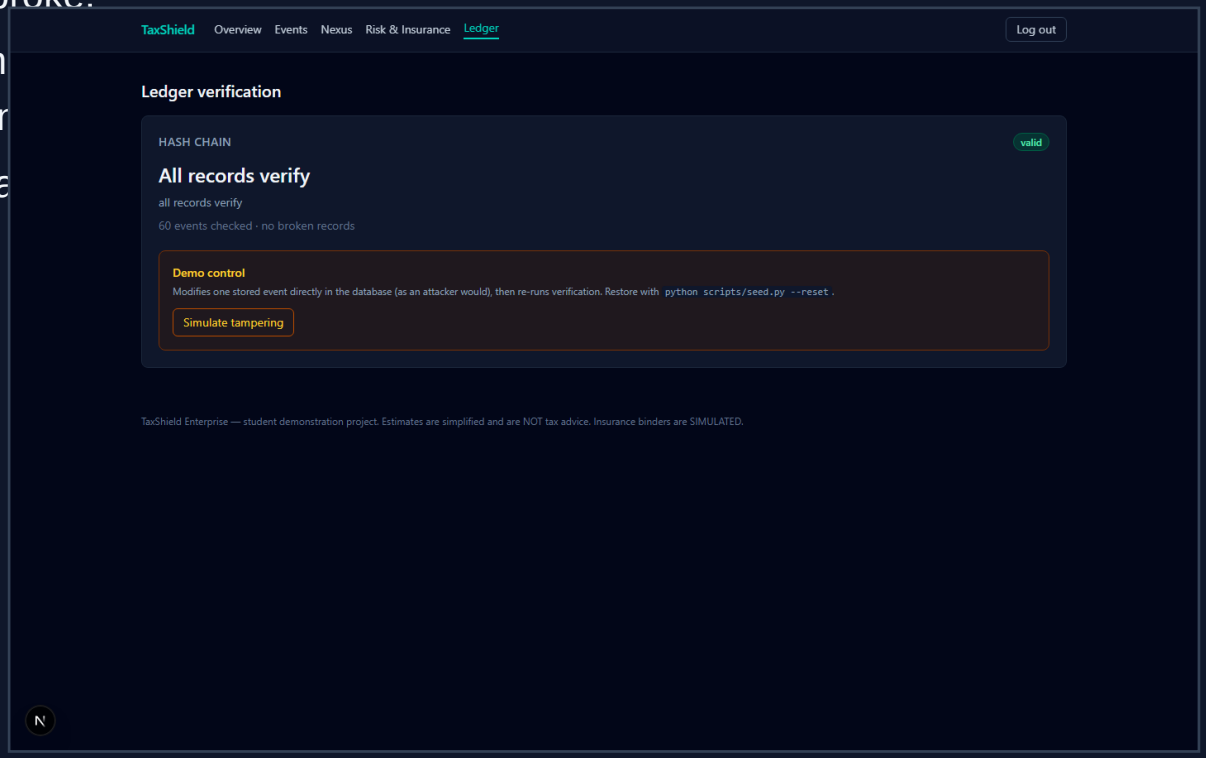
Measured on 48 hand-labeled events (real pipeline, no cherry-picking):

Engine	Accuracy	Precision	Recall	F1
keyword-rules (default, offline)	97.9%			
llm-rag	not measured — needs a free-tier API key; harness ready			

The single miss: a hosted-model consumption event ('Bedrock model invocation') triggered on the word *model*. We publish the miss because judges should trust measured numbers over claimed ones.

Security & audit trail

- **AES-256-GCM** per event, fresh 12-byte nonce, key from environment only.
- **SHA-256 hash chain** over ciphertexts: edit or reorder one row and verification reports *first_broken_seq* — the exact record that broke.
- **JWT (HS256)** on every data route; rate lim
  to the dashboard origin; Pydantic validatior
- **21 passing pytest tests** incl. wrong-key fa
  pip-audit clean.



Stated limits (SECURITY.md): single non-rotating key; chain proves tampering but cannot stop a full rewrite by a key-holder; no enclaves/mTLS.

Market size (assumptions visible)

TAM	**$26.39B**	research-credit prep market — Gate 1 secondary sources, directional
SAM	**$5.20B**	US tech firms <$50M ARR addressable with self-serve telemetry
SOM	**$217.5M**	15,000 firms × $14,500 ACV (firm count is an assumption)

Year-1 target: 80–120 customers ≈ $1.2–1.7M ARR. Every dollar is arbitrage: the customer pays $14.5k for an estimated $50k–$250k credit (their numbers, their CPA's sign-off).

TAM/SAM carried from Gate 1 and not re-verified this gate; SOM arithmetic shown so it can be challenged.

Competition & differentiation

Player	Does	Misses
CPA firms	Expert credit calc	Per-engagement pricing; manual evidence chase
Avalara / tax engines	Sales tax & nexus	Nexus yes — no R&D evidence pipeline
TurboTax Business	Form filling	Forms, not evidence; user must know what qualifies
AI tax assistants	Chat answers	Generic knowledge; not wired to your telemetry; no audit tra
TaxShield	**Telemetry → claim**	**Credit as a by-product of data you already g audit-trail wrapped**

Business model & go-to-market

- **Monitor — $499/mo:** telemetry ingest, classification, ledger, nexus watch (<25 employees).
- **Claim — $14,500/yr (core):** + CPA-ready export package, audit-trail walkthrough, annual recompute.
- **Enterprise — custom:** SSO, Postgres/on-prem, enclave processing, multi-entity.

Unit economics (illustrative): >85% gross margin at Tier 2 — LLM cost <$0.02/event (keyword fallback = $0), hosting ~$50/mo per customer band; 6-month payback at $14.5k ACV if close rate ≥20%.

Go-to-market

- **CPA channel first:** firms doing credits for 10+ clients — sell evidence quality, they keep the filing relationship.
- **Founder-led content:** 'what your Git history says about §41'.
- **Incubator partnerships:** portfolio firms form habits early.
- **Free nexus watch** as top-of-funnel.

Roadmap, team & ask

- Run 3–5 customer interviews; revise the business case with real evidence.
- Wire a free-tier LLM key; measure llm-rag accuracy on the same 48 labels.
- Verify the four unverified state thresholds from official sites.
- Deploy for a public demo link; Postgres migration.
- Timesheet integration to replace the $75/hr hours assumption.
- Enclaves/mTLS for enterprise; underwriting partner for real insurance.

Team Qubrix

AI/ML · Cyber-Security · IoT & Cloud

Ask

Mentor intro to a CPA firm running §41 credits for 10+ clients (channel test), and feedback on the honesty of our real/partial/mocked claims.

Demo: python run.py → dashboard on :3000 · everything runs offline · full evidence in github.com/Thiviru-Opallage/taxshield-enterprise